

Finite Blaschke Products Answer the Existence Part of a Toeplitz Problem on $H^1(\mathbb{C}^+)$

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Status and source question

Status: substantial partial solution, likely valid, pending human review.

Bellavita and Peloso [1] study, for a nonconstant inner function Θ on the upper half-plane, the codimension-one subspace

$$H_\Theta^1 = \left\{ f \in H^1(\mathbb{C}^+) : \langle f, \Theta \rangle := \int_{\mathbb{R}} f(x) \overline{\Theta(x)} dx = 0 \right\}$$

and the restricted Toeplitz operator $T_\Theta : H_\Theta^1 \rightarrow H^1(\mathbb{C}^+)$. Their Question 1 on PDF page 12 asks whether any inner functions other than $e^{i\tau z}$ yield a bounded restricted operator and, more broadly, which inner functions do. Their Theorem 1.3 shows that boundedness is equivalent to

$$H_\Theta^1 = K_\Theta^1 \oplus \Theta H^1, \quad K_\Theta^1 = \{f \in H^1(\mathbb{C}^+) : \overline{\Theta} f \in \overline{H^1(\mathbb{C}^+)}\}. \quad (1)$$

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4. OPEN QUESTIONS

Question 1. Clearly, the first question we are interested in is whether there are inner functions Θ , other than the meromorphic inner functions $e^{i\tau(\cdot)}$, such that the associated Toeplitz operator $T_\Theta : H_\Theta^1 \rightarrow H^1(\mathbb{C}^+)$ is bounded, or equivalently, for which inner functions Θ the equality

$$H_\Theta^1 = K_\Theta^1 \oplus \Theta H^1$$

holds. Moreover, from Theorem 1.3 we know that, if Θ is an inner function, $T_\Theta : H_\Theta^1 \rightarrow H^1(\mathbb{C}^+)$ is unbounded if and only if

$$K_\Theta^1 \oplus \Theta H^1 \subsetneq H_\Theta^1,$$

that is, if and only if the quotient $H_\Theta^1 / (K_\Theta^1 \oplus \Theta H^1)$ is non-trivial. When non-trivial, it would be interesting to describe it.

Question 2. The second question delves deeper into the subject and it is related to another interesting problem. In analogy with the holomorphic case, we introduce a subspace of the real variable Hardy space $H^1(\mathbb{R})$. For $b \in \text{BMO}(\mathbb{R})$ we consider the pre-annihilator of $\{b\}$,

Figure 1: Question 1 on source PDF page 12.

Proof intuition

For a finite Blaschke product B of degree N , the quotient H^1/BH^1 is just the N -dimensional space of jets at the zeros of B . The single annihilation condition defining H_B^1 cuts this quotient down to dimension $N - 1$.

At the endpoint $p = 1$, one must take care not to import an unavailable Riesz projection. Instead, the model space can be computed directly. If Q is the polynomial whose roots are the reflected zeros of B , then analytic and anti-analytic Fourier support force every $f \in K_B^1$ to have the form P/Q . Boundary integrability is exactly the condition $\deg P \leq N - 2$. Thus K_B^1 also has dimension $N - 1$. The source's elementary inclusion $K_B^1 \oplus BH^1 \subseteq H_B^1$ must therefore be an equality.

The finite-Blaschke theorem

Theorem 1. *Let*

$$B(z) = \eta \prod_{j=1}^s \left(\frac{z - a_j}{z - \bar{a}_j} \right)^{m_j}, \quad a_j \in \mathbb{C}^+, \quad |\eta| = 1,$$

be a nonconstant finite Blaschke product, with distinct a_j , positive multiplicities m_j , and degree $N = \sum_j m_j$. Then

$$H_B^1 = K_B^1 \oplus BH^1.$$

Consequently $T_{\bar{B}} : H_B^1 \rightarrow H^1(\mathbb{C}^+)$ is bounded. If $f = k + Bh$ is the displayed decomposition, its bounded extension satisfies $T_{\bar{B}}f = h$.

Proof. We divide the proof into three short steps.

Step 1: the jet quotient. Consider the bounded jet map

$$J : H^1(\mathbb{C}^+) \longrightarrow \mathbb{C}^N, \quad Jf = (f^{(r)}(a_j))_{1 \leq j \leq s, 0 \leq r < m_j}.$$

It is onto. Indeed, fix $M > N + 1$. Hermite interpolation applied to functions of the form $P(z)/(z + i)^M$, with $\deg P < N$, realizes arbitrary prescribed jets; these rational functions belong to $H^1(\mathbb{C}^+)$.

The kernel of J is BH^1 . One inclusion is immediate. Conversely, if all the prescribed jets of f vanish, then f/B is analytic in $\mathbb{C}^+$ and has the same boundary modulus as f . The standard inner-outer factorization (equivalently, the Smirnov theorem) gives $f/B \in H^1(\mathbb{C}^+)$. Hence

$$H^1(\mathbb{C}^+)/BH^1 \cong \mathbb{C}^N. \tag{2}$$

Let $L_B(f) = \langle f, B \rangle$. Since every $h \in H^1(\mathbb{C}^+)$ has integral zero,

$$L_B(Bh) = \int_{\mathbb{R}} h(x) dx = 0,$$

so $BH^1 \subseteq H_B^1 = \ker L_B$. The functional L_B is nonzero: under the H^1 -BMOA duality, it is represented by the nonconstant function B , which is nonzero modulo constants. It follows from (2) that

$$\dim(H_B^1/BH^1) = N - 1. \tag{3}$$

Step 2: the endpoint model space. Put

$$A(z) = \prod_{j=1}^s (z - a_j)^{m_j}, \quad Q(z) = \prod_{j=1}^s (z - \bar{a}_j)^{m_j}, \quad B = \eta A/Q.$$

We claim that

$$K_B^1 = \left\{ \frac{P}{Q} : P \text{ is a polynomial and } \deg P \leq N - 2 \right\}. \quad (4)$$

Let $f \in K_B^1$. There is a $g \in H^1(\mathbb{C}^+)$ such that $\overline{B}f = \overline{g}$ on $\mathbb{R}$, and hence

$$\overline{\eta} Qf = A\overline{g}$$

as tempered distributions on the boundary. The Fourier transform of Qf is supported in $[0, \infty)$, while that of $A\overline{g}$ is supported in $(-\infty, 0]$. Their common Fourier transform is therefore supported at $\{0\}$. A tempered distribution with point support is a finite linear combination of derivatives of the Dirac mass, so $Qf = P$ is a polynomial. Since $f = P/Q$ belongs to $L^1(\mathbb{R})$ and $\deg Q = N$, necessarily $\deg P \leq N - 2$.

Conversely, take such a polynomial P and put $P^\#(z) = \overline{P(\overline{z})}$. Both

$$f = P/Q, \quad g = \eta P^\# / Q$$

belong to $H^1(\mathbb{C}^+)$, and on $\mathbb{R}$ one has $\overline{B}f = \overline{g}$. Thus $f \in K_B^1$, proving (4). In particular,

$$\dim K_B^1 = N - 1, \quad (5)$$

with $K_B^1 = \{0\}$ when $N = 1$.

Step 3: fill the hyperplane. If $k \in K_B^1$, write $\overline{B}k = \overline{g}$ with $g \in H^1(\mathbb{C}^+)$. Then

$$\langle k, B \rangle = \int_{\mathbb{R}} \overline{g(x)} dx = 0,$$

so $K_B^1 \subseteq H_B^1$. Also $K_B^1 \cap BH^1 = \{0\}$: after cancelling B , a boundary function lying in both H^1 and $\overline{H^1}$ must be zero. Hence K_B^1 injects into H_B^1/BH^1 . Equations (3) and (5) show that this injection is onto. Therefore $H_B^1 = K_B^1 \oplus BH^1$.

The boundedness of $T_{\overline{B}}$ now follows from (1). On K_B^1 the Toeplitz operator vanishes, while $T_{\overline{B}}(Bh) = h$, which gives the asserted action. $\square$

Corollary 2. *Every nonconstant rational inner function on $\mathbb{C}^+$ gives a bounded restricted Toeplitz operator. In particular,*

$$B(z) = \frac{z - i}{z + i}$$

answers the existence part of Question 1 affirmatively; here $K_B^1 = \{0\}$ and $H_B^1 = BH^1$.

Verification, scope, and novelty

The proof treats repeated zeros through the full Hermite jet family. It also checks the endpoint feature that a possible $1/x$ rational tail is not in L^1 ; this is precisely why the numerator degree in (4) is $N - 2$, rather than the familiar $N - 1$ from K_B^2 . Since BH^1 is closed and K_B^1 is finite-dimensional, the algebraic direct sum is automatically topological, so its coordinate projections and the resulting Toeplitz extension are bounded.

This is a partial answer to the source's classification problem. It settles the existence question and the full rational-inner subcase, but it does not classify infinite Blaschke products, singular inner functions, or general meromorphic inner functions.

On 9 August 2026 we searched the run's registry, solution, attempt, and proof-gap indexes for arXiv:2503.07281 and the core terms. Bounded web searches used the exact paper title, the notation H_Θ^1 and K_Θ^1 , the displayed equality, finite Blaschke products, and later work by the source authors. They found the source paper and general model-space background, but no separate work recording this finite-Blaschke answer. This supports, but does not guarantee, novelty.

Human-review focus

The decisive checks are: (i) the endpoint identity $\ker J = BH^1$; (ii) the Fourier-support argument proving (4); and (iii) the fact that the nonconstant B defines a nonzero H^1 functional, producing exactly one dimension of codimension inside the jet quotient.

References

- [1] C. Bellavita and M. M. Peloso, *On Toeplitz operators on $H^1(\mathbb{C}^+)$* , arXiv:2503.07281v1 (2025), accepted in Proc. Amer. Math. Soc.
- [2] J. B. Garnett, *Bounded Analytic Functions*, Graduate Texts in Mathematics, Springer, 2006.